FRANKLIN CRYPTO RESEARCH COMPETITION · UBC 2026

Solana (SOL) Buying Reliability Before It Is Priced

An engineering thesis on the settlement layer of onchain finance — and the Alpenglow re-rating the market has not yet made.

LONG · Track 02 — Market Cap Over $250M

TEAM	UNIVERSITY / SOCIETY	CONTACT	DATE
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Price & fundamentals as of early November 2026. Primary data: Blockworks Research & Allium (competition toolkit), Anza, Messari, 21Shares, DeFiLlama. Independent student research; not investment advice.

PRICE	CIRC. MKT CAP	ANN. REV	ANN. APP REV	STABLECOINS	BASE TARGET
~$89	**~$53B**	**~$303M**	**~$1.2B**	**~$16.8B**	**$160**
–69% from ATH $293	Rank #6–7	**+48% QoQ (Q3'26)**	#1 chain, 9 qtrs	All-time high	**+80% / 12–18mo**

Recommendation: LONG SOL at ~$89 (~$53B circulating cap). 12–18 month targets — **Bear $55 (–38%)**, **Base $160 (+80%)**, **Bull $290 (+226%)**. The payoff is asymmetric because the market still prices Solana on the two things that are weak right now — memecoin-era throughput and ETF flows — and has not repriced the one thing that structurally changed the asset: Solana now has the best-proven reliability of any high-throughput chain, and it earned that title three weeks before this competition.

THE VARIANT PERCEPTION

Consensus treats "fast and cheap" as Solana's pitch — but that is commoditized; every new L1 and L2 is fast and cheap. Our claim is non-obvious: **in a world of abundant blockspace, the scarce, value-accruing resource is execution quality** — uptime, deterministic sub-second finality, low landed-transaction failure, predictable fees. The market already pays a large, explicit premium for it: Hyperliquid (HYPE) — a Long >$250M finalist in this competition last cycle — trades at **~27× protocol revenue** precisely because it is fast, never down, and routes ~99% of fees to buybacks. On **October 21, 2026**, Solana swapped its entire consensus engine — the 18-month **Alpenglow** upgrade — in a **single slot with zero downtime** (a first for any live chain), cutting finality from ~12.8s to a ~150ms target under a hardened "20+20" fault model. That converts its biggest historical liability — seven outages through early 2024 — into a measurable strength: **~30 months of 100% uptime.** Alpenglow lets a $53B base layer earn the execution re-rating a single app already enjoys, at a scale no app can match.

WHY THE FUNDAMENTALS FIT — AND THE OBJECTION WE ANSWER FIRST

Solana is **#1 in application revenue for nine straight quarters** (~$299M in Q3 2026, +48% QoQ, ~$1.2B annualized — Blockworks), #1 in DEX volume, and sits on an all-time-high **~$16.8B stablecoin base**; network REV just inflected +48% QoQ, its first rise in six quarters. We will not pretend SOL is statistically cheap — on raw network REV it trades near **175×**. But that denominator is broken for every base layer in abundant blockspace (Ethereum screens above 1,000×); the right lenses — app revenue (~44×, re-accelerating) and the execution re-rating — point the other way. **You are buying a mispriced factor at a 69% drawdown, not a cheap token.**

One-line thesis. The market pays ~27× revenue for execution quality (HYPE); Alpenglow makes Solana the only $50B-scale base layer that can credibly claim it; the re-rating is not yet priced, and the catalyst is already live.

WHY NOW (CATALYSTS)

- **Alpenglow live** (Oct 21) — reliability re-rating not yet in price
- **Firedancer** stake ramp → client-diversity fix
- **Staking-enabled spot ETFs** — Franklin SOEZ, Bitwise BSOL
- **Tokenized RWA** — Franklin's BENJI settles on Solana; Rotor next

KEY RISKS (ADDRESSED P.4)

- Rich on *network* REV → thesis leans on app-rev + re-rating
- Client concentration (Nakamoto 1) — Firedancer is the fix
- Geographic stake concentration (~59% DE/NL/US)
- Near-term price/ETF-flow weakness; 150ms/Rotor execution risk

1 What Solana is now: from "fast chain" to settlement layer

Solana is a single-state, high-throughput Layer-1 that settles transactions in one global execution environment rather than fragmenting liquidity across rollups. For most of its life the bull case was performance and the bear case was reliability. In 2026 that framing is obsolete. Throughput is no longer scarce — it is a race to zero across dozens of chains. What is scarce is a venue that institutions and high-value applications can trust to be *always on, fast to finality, and cheap to verify*. Delphi Digital frames Solana's 2026 upgrade cycle as building the "decentralized Nasdaq" — an exchange-grade environment where an onchain central-limit order book can compete with a centralized exchange on latency and fairness. That is the lens that matters, and it is the lens this panel's institutions — a tokenized-fund issuer (Franklin Templeton), a financial-infrastructure fund (Collab+Currency), and the research shop that invented the REV metric (Blockworks) — already use.

2 The engineering turn: Alpenglow (the core of this thesis)

Alpenglow is the largest change to Solana since launch, and the detail matters because the thesis is an engineering one. It replaces TowerBFT with two components. **Votor** is a new voting and finality path using BLS signature aggregation and a two-round fast/slow design: a block finalizes in one round at ≥80% stake participation and two rounds at ≥60%. Votes move *off-chain*, which both collapses finality (from ~12.8s to a ~150ms target) and frees an estimated 70%+ of block space previously consumed by vote transactions. **Rotor** (the propagation layer that succeeds Turbine) is the next stage and is not yet live — we treat it as upside, not as a claim. The economic redesign is underappreciated: vote fees are replaced by a burned ~0.8 SOL/epoch "validator admission ticket," cutting validator operating costs by roughly 60% and making the validator set cheaper to join and more sustainable to run.

The resilience upgrade is the part institutions will care about: Alpenglow's **"20+20"** model keeps finalizing through 20% actively malicious stake *and* 20% crashed/offline stake at the same time, a materially stronger guarantee than TowerBFT's <33% single-fault threshold. And the migration itself is the proof point — Anza executed an **18-month build as a one-slot "hot swap" on live mainnet with zero downtime**, the first time any network has replaced its consensus engine without halting. It was rehearsed across nine at-scale migrations and 1,500-node simulations and shipped with a reversible controlled-pause window. This is not a roadmap promise a judge has to take on faith; it is a dated, verifiable engineering milestone.

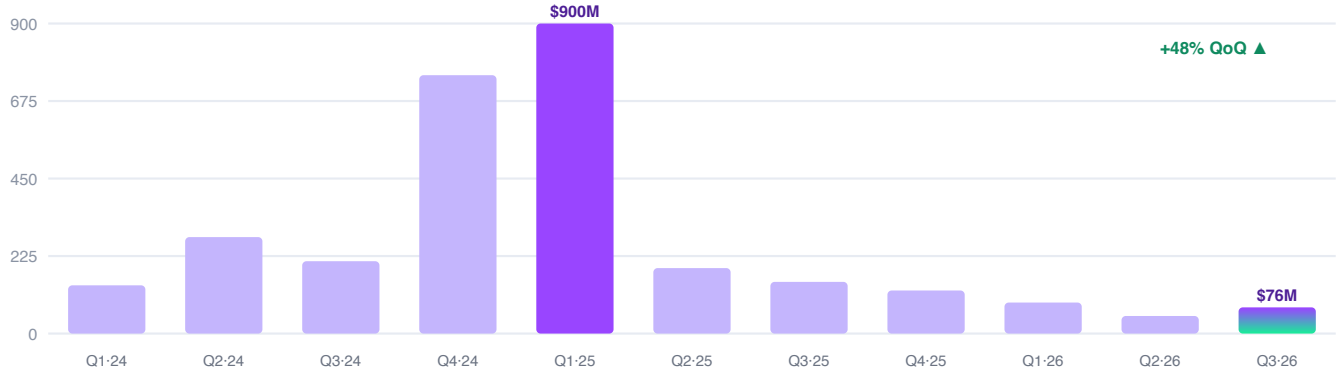
> **THE WEAKNESS-TO-STRENGTH PIVOT (THE WHOLE ARGUMENT IN FOUR NUMBERS)**
>
> **7** full-network halts between Dec 2020 and Feb 2024 · **Feb 6, 2024** last outage (~5 hours) · **~30 months** of 100% uptime since · **~12.8s** → **~150ms** finality via a zero-downtime migration. The market still prices the first number. We are underwriting the last three.

3 Market analysis & traction: the economic engine

Strip out the narrative and Solana is, by the metrics this competition is judged on, the most economically active base layer in crypto. It has led all chains in **application revenue for nine consecutive quarters**, printed **~$299M in Q3 2026 (+48% QoQ)**, and carries the deepest onchain stablecoin base outside Ethereum and Tron at an all-time-high **~$16.8B**. Its network REV — Blockworks' fees-plus-MEV metric — collapsed ~87% as the 2024–25 memecoin cycle unwound, and then *inflected +48% QoQ in Q3 2026*, the first increase in six quarters. That inflection, off a washed-out base, is the fundamental tell that the franchise is broadening beyond speculation into payments, DeFi, and tokenized assets.

Figure 1 — Solana network REV: the memecoin boom, the washout, and the Q3'26 inflection

Quarterly real economic value (transaction fees + MEV tips), USD millions. Source: Blockworks Research.



The revenue is diversified across real applications, not one casino: Jupiter (the dominant swap router), Raydium and the broader DEX complex (Solana has out-traded Ethereum in monthly DEX volume since late 2024), Pump.fun on the consumer side, plus fast-growing stablecoin payments and tokenized-equity volume. Daily active addresses average ~2.3M. The point is not that any single app is irreplaceable; it is that demand is *broad*, which is exactly what makes a base-layer bet more durable than a single-app bet like HYPE.

4 Token assessment & value accrual

SOL has ~590M circulating of ~636M total supply. Issuance is ~3.6% and disinflating 15%/yr toward a 1.5% floor; ~68–69% of supply is staked (~$38B of economic security), earning a ~5–6% gross yield. Value accrues to the token three ways: (i) base + priority fees and MEV paid to stakers (REV), (ii) a protocol burn on fees, and (iii) the staking yield itself, now net-positive in real terms for most validators post-Alpenglow cost cuts.

Demand is structurally new. Spot ETFs are live and *staking-enabled* — Franklin's SOEZ (0.19%) and Bitwise's BSOL among them — holding ~$1.3B+ and growing, with more issuers filed. Digital-asset treasury companies have accumulated tens of millions of SOL. Against ~3.6% issuance, ETF + staking + treasury lockup is removing float faster than the chain is minting it — a supply/demand setup the memecoin-era price does not reflect.

5 Catalysts & risks

CATALYSTS (NEXT 3–12 MONTHS)

- **Reliability re-rating** as Alpenglow's zero-outage record compounds post-migration
- **Firedancer** stake share rising → fixes the client-diversity knock
- **ETF inflows** turning; staking yield makes SOL ETFs structurally attractive vs BTC/ETH
- **Tokenized equities & RWA** — Franklin BENJI already settles on Solana; CLOB/HFT use cases unlocked by 150ms finality
- **Rotor** ships → bandwidth + further decentralization

RISKS & MITIGANTS

- **Rich on network REV (~175×).** Mitigant: wrong denominator in abundant blockspace; app-rev ~44× and re-accelerating
- **Client concentration (Nakamoto 1).** Mitigant: Firedancer is live and climbing — a dated fix, not a hope
- **Geographic stake concentration (~59% DE/NL/US).** A genuine open risk we flag honestly
- **REV cyclicality / ETF outflows / price weakness.** Mitigant: entry at –69% already discounts this
- **Execution risk** on sustained 150ms & Rotor. Mitigant: thesis needs the step-change, not the asymptote

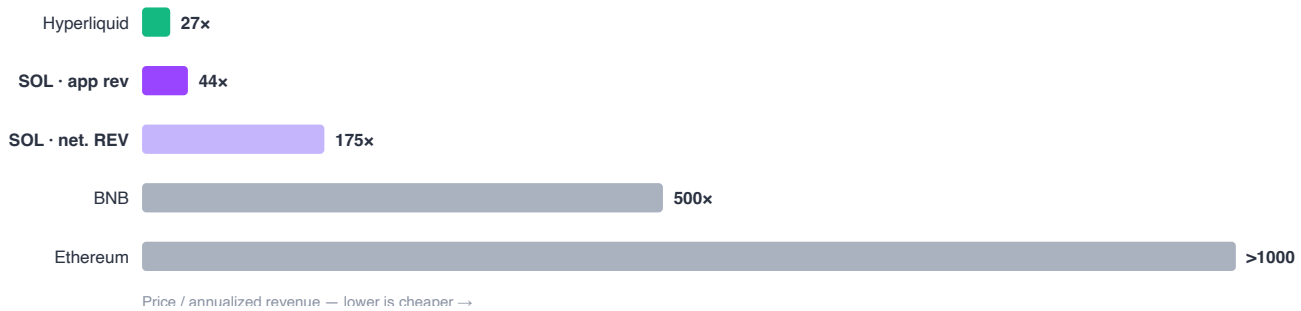
Position: Long SOL. **Entry:** ~$89 (~$53B cap). **Horizon:** 12–18 months. **Exit/target (base):** $160. The thesis resolves as the market re-rates Solana from a memecoin-beta L1 to an execution-grade settlement layer, while app revenue compounds and net float shrinks.

VALUATION — THREE LENSES, ONE CONCLUSION

(1) Reframe the multiple. We concede SOL is not cheap on network REV (~175×). But that metric has broken down for every base layer as blockspace became abundant — Ethereum screens above 1,000× on the same denominator. The economically meaningful multiple is price-to-*app revenue*, where Solana trades near **44×** on ~$1.2B annualized and is *growing* (+48% QoQ), versus a shrinking base elsewhere. **(2) Price the factor.** The market pays ~27× protocol revenue for Hyperliquid's execution quality; Alpenglow qualifies Solana for the same factor at base-layer scale and diversification. **(3) Supply/demand.** Staking-enabled ETFs, DAT accumulation, and ~68% staked lockup are removing float faster than ~3.6% issuance adds it.

Figure 2 — The re-rating, visualized: what the market pays for execution quality

Price-to-annualized-revenue by chain. Network REV for L1s; protocol (app) revenue for Hyperliquid. Source: Blockworks Research; DeFiLlama. Nov 2026.



SCENARIO PRICE TARGETS (12–18 MONTHS)

SCENARIO	SOL	IMPLIED CAP	RETURN	WHAT HAS TO BE TRUE
Bear	$55	~$33B	**–38%**	REV recovery stalls; ETF outflows persist; Firedancer/decentralization slips; risk-off tape.
Base	**$160**	**~$95B**	**+80%**	**App revenue compounds ~20–30%; reliability re-rating begins; ETF AUM grows; modest multiple expansion.**
Bull	$290	~$172B	**+226%**	Solana becomes default settlement venue for stablecoins & tokenized assets; execution factor fully priced; ETF+DAT demand overwhelms issuance; ATH retest.

Probability-weighted (25% / 50% / 25%) expected value ≈ $166, or ~+87% from ~$89 — a positively skewed payoff with a drawdown already discounted by the 69% decline from the prior high.

Why this is the trade, not just the token. The consensus SOL pitch (price, TPS, ETF flows) is weak right now, which is the opportunity: the market is looking at the wrong, lagging variables. Reliability and execution quality are the variables that re-rate assets — the market proved it with Hyperliquid — and they inflected for Solana on October 21. We are positioned in front of the re-rating, not behind it.

Conclusion

Solana spent four years being underwritten on the thing it did worst — staying online. On October 21, 2026 it retired that liability in a single slot, and in doing so it quietly changed what kind of asset it is: from a high-beta throughput bet into the only $50B-scale base layer that can credibly claim exchange-grade reliability and sub-second finality. The market has not repriced that, because it is still watching memecoin REV and ETF flows — both near cyclical lows — instead of the variable that actually drives re-ratings.

We know the market pays for execution quality because it already does: Hyperliquid earns ~27× revenue for being fast and never down. Alpenglow makes the same case for Solana, but on a franchise with nine straight quarters of #1 app revenue, an all-time-high stablecoin base, a freshly inflecting REV line, and structural ETF and staking demand shrinking its float. Buying at ~$89 — 69% below the prior high, three weeks after the catalyst — is buying a mispriced factor with the downside already discounted.

Recommendation: Long SOL, 12–18 months, base target $160 (+80%), bull $290 (+226%), bear $55 (–38%). The reliability is proven. The re-rating is not yet priced. That gap is the trade.

A Appendix — valuation detail & data notes

VALUATION COMPARABLES (Nov 2026)

CHAIN	CAP	ANN. REV	P/REV	STBL.
Solana	**$53B**	**$0.30B**	**~175×**	**$16.8B**
Ethereum	$280B	~$0.26B	>1,000×	$137B
BNB Chain	$95B	~$0.18B	~500×	$13.8B
TRON	$31B	~$0.30B	~100×	$89B
Hyperliquid*	~$19B	~$0.70B	~27×*	$6.4B

*Hyperliquid P/REV uses protocol (perp-trading) revenue, the metric its token captures via buybacks — the correct lens for an app-chain and the execution-quality comp. L1 rows use network REV (fees+MEV). SOL app-rev basis ≈ 44×. Source: Blockworks Research; DeFiLlama.

ALPENGLOW — TECHNICAL SUMMARY

DIMENSION	BEFORE	AFTER
Consensus	TowerBFT	Votor (BLS)
Finality	~12.8s	~150ms target
Fault model	<33%	20% + 20%
Votes	On-chain	Off-chain
Validator cost	Baseline	~–60%
Uptime (30mo)	7 halts →	100%

Live on mainnet-beta Oct 21, 2026 (epoch 1064, slot 459,653,000). Rotor (propagation) pending — treated as upside. Live finality measured ~430ms (Messari, Oct 2026). Source: Anza; Messari; Helius.

Data note: network REV, app revenue, stablecoin supply, DEX volume and market cap are drawn from the Blockworks Research "Blockchains" dataset (the competition toolkit; REV = base + priority fees + MEV tips) and cross-checked against Blockworks' Solana Q3 2026 Token Holder Report and 21Shares. Annualized figures use the Q3 2026 run-rate ×4. Price/cap as of early Nov 2026.

R References

1. Blockworks Research — Blockchains dataset & Solana Q3 2026 Token Holder Report (app.blockworks.com).
2. Blockworks Research — "REV: Real Economic Value" methodology.
3. Anza — Alpenglow upgrade & mainnet activation (anza.xyz/alpenglow; solana.com/upgrades/alpenglow).
4. Messari — Solana October 2026 update; measured finality.
5. Helius / Syndica — Alpenglow technical explainers (Votor, Rotor, 20+20).
6. 21Shares — Solana H1 2026 network revenue (–87% YoY).
7. DeFiLlama — Hyperliquid fees/revenue; cross-chain DEX volume & TVL.
8. Delphi Digital — "Solana as a decentralized Nasdaq" (Jan 2026).
9. Franklin Templeton — Franklin Solana ETF (SOEZ); BENJI / FOBXX on Solana.
10. Bitwise — Solana staking report (Q3 2026); BSOL spot ETF.
11. Anza/Solana Foundation — SIMD-0326 governance vote; validator economics.
12. CoinDesk / The Block — Hyperliquid revenue, buybacks & perp market share (2026).